9.12 Random Subsets

WORKSHEET

The question

STEP III 2019, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself How can you describe a subset of $\{1, \ldots, n\}$ by a sequence of n answers to yes/no questions?

STAGE 2 Disjoint sets

Ask yourself For $A_1 \cap A_2 = \varnothing$, what must be true of each element i? What is the probability of that, for one element? Why can you multiply over the elements?

STAGE 3 Subsets

Ask yourself For $A_1 \subseteq A_2$, which of the four possibilities for an element is forbidden?

Ask yourself For $A_1 \subseteq A_2 \subseteq A_3$, which of the 8 patterns of membership (in or out of each set) are allowed for an element?

Try this next

1 **Covering.** Find $\mathrm{P}(A_1 \cup A_2 = S)$ and $\mathrm{P}(A_1 \cup A_2 \cup \cdots \cup A_m = S)$.

2 **Same size.** Show that $\mathrm{P}(|A_1| = |A_2|) = \binom{2n}{n}/4^n$. (*Hint:* $|A_1| = |A_2|$ exactly when $|A_1| + |S \setminus A_2| = n$.) Show that this is about $1/\sqrt{\pi n}$ for large n.

3 **Partitions into three.** Show that the number of ordered pairs of disjoint subsets (A, B) of S is 3^n, and find the number of unordered pairs $\{A, B\}$ of disjoint subsets.

4 **Sperner for small n.** For $n = 3$ and $n = 4$, find the largest family of subsets of $\{1, \ldots, n\}$ in which no set contains another.

Further reading

- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific), the chapters on subsets and partially ordered sets.
- **Béla Bollobás, *Combinatorics: Set Systems, Hypergraphs, Families of Vectors and Combinatorial Probability*** (Cambridge University Press). A short, elegant book on families of subsets, from Sperner's theorem onwards.

- **Peter Cameron, *Combinatorics: Topics, Techniques, Algorithms*** (Cambridge University Press). A first course with a strong emphasis on counting in two ways.

The full guided solution, and what the question leads to at university, is in Section 9.12:
https://beyondstep.keithmcnulty.org/ch09/random-subsets.html